Week 07: Proposal Writing II: Evaluation, Reviewers and Budget Justification

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Proposals are judged by people under time pressure, and their judgements vary. This week examines how proposals are evaluated, what research shows about the reliability of peer review, how to write for reviewers and how to justify a budget. A mock panel lets students take the reviewer's seat [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to describe the stages of proposal evaluation. They should be able to explain the evidence on reviewer disagreement and its consequences near the funding line. Finally, they should be able to identify common weaknesses of proposals and to justify a budget in relation to tasks.

Lecture

How proposals are evaluated

Evaluation usually proceeds in stages. Experts first review proposals individually against the criteria of the call and then reach consensus in a group or panel. The funder ranks the proposals and funds as many as the budget allows. In Horizon Europe, independent experts score each proposal on excellence, impact and implementation, with thresholds and ranking rules that Week 5 introduced [1]. TÜBİTAK's research programmes rely on panels and reviewers who assess originality, method, project management and wider impact with equal weight [4]. The ERC uses a single criterion, scientific excellence, applied to both the project and the principal investigator.

How reliable is peer review?

Peer review of proposals is less reliable than applicants often assume. Pier and colleagues asked experienced reviewers to evaluate the same grant applications of the National Institutes of

Health [2]. They found low agreement between reviewers, both in their scores and in their written critiques. Graves, Barnett and Clarke analysed scores by members of a grant review panel and showed that the variation between reviewers could change funding decisions for proposals near the funding line [3]. Figure 7.1 illustrates the mechanism with a simple model. When reviewers' scores contain noise, the better of two similar proposals is ranked higher only with a modest probability. Adding reviewers helps but does not remove the effect. The interactive lab simulates whole panels and shows how much the set of funded proposals can change between two panels that see the same proposals.

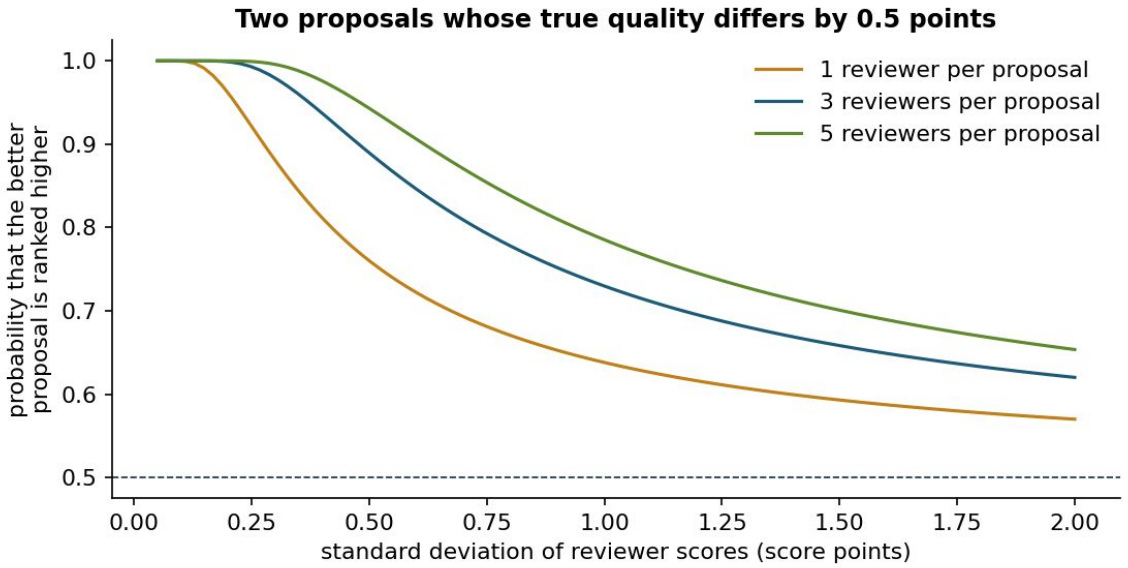


Figure 7.1. Probability that the better of two proposals is ranked higher when their true quality differs by half a point. The probability depends on reviewer noise and on the number of reviewers per proposal in a simple normal model.

Writing for reviewers

The practical conclusion is to reduce the room for misreading. Reviewers look for the answer to each criterion, so the proposal should make it easy to find. Headings follow the criteria, and objectives are stated once and referred to consistently. The method addresses every objective, preliminary results make feasibility credible and the effort in the work plan matches the ambition. Common weaknesses include vague objectives, a method that does not match the objectives, missing risk analysis, impact claims without users or measures, and budgets that are not linked to tasks. The interactive lab asks students to sort such reviewer comments by criterion, which trains the habit of reading one's own draft as a reviewer would.

Budget justification

A budget is part of the argument. Each cost should follow from a task. Person-months follow from the effort of work packages, and equipment from a method that needs it and cannot use existing infrastructure. Travel follows from dissemination and collaboration, and services from tasks that the team cannot do itself. Funders define eligible cost categories and limits in their

rules, and Horizon Europe adds a flat rate for indirect costs to eligible direct costs [5]. Reviewers are sensitive to both overbudgeting, which signals poor planning, and underbudgeting, which signals that the work cannot be done as described.

> ***Pause and reflect.*** *In the mock panel, how did your scores compare with the scores you would expect for your own proposal? What does the difference suggest?*

Hands-on components

Interactive lab: Part A asks you to score three short proposal summaries, written for this exercise, on the three Horizon Europe criteria. Part B simulates panels with noisy reviewers. Part C sorts typical reviewer comments by criterion [1, 2, 3]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-07/lab.html

Colab notebook: It models reviewer noise, computes the probability that the better of two proposals is ranked higher and simulates the agreement between two independent panels [2, 3]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-07/NB07_proposal_writing_2.ipynb

Reflection prompts

1. Which of the three summaries in the mock panel did you rank first, and which comment would you write to its authors?
2. Given the evidence on reviewer disagreement, how should an applicant respond to a rejection with good scores?
3. Which item of your draft budget would a reviewer question first, and how will you justify it?

Weekly task and submission

What to do. This task has two parts. First, review one anonymised proposal draft of another student, or a public example, with the official criteria of its target programme. For each criterion, give a score and a comment of about 150 words. Second, write the budget justification of your own proposal and link every item to the tasks that need it.

Where to look. Lecture: sections "How proposals are evaluated", "Writing for reviewers" and "Budget justification". Interactive lab: Part A for scoring three summaries and Part C for the criterion that each comment concerns. Colab notebook: Section 1 with Exercise 1, which shows why noisy reviewers make clear writing important near the funding line.

Example. A useful comment names a weakness and its consequence: The method does not explain how the labelled data will be obtained, so the second objective may not be feasible. A budget line could read: 12 person-months of a research assistant for Work Package 2, which builds and tests the detection model.

What to submit. The review with its scores and comments, the budget justification, and the Colab notebook with Exercises 1 and 2 completed. The midterm capstone, which also ends this week, is the complete proposal and may build on this task.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] European Commission (2021). Standard briefing slides for experts: Horizon Europe. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/experts/standard-briefing-slides-for-experts_he_en.pdf

[2] Pier, E. L., Brauer, M., Filut, A., Kaatz, A., Raclaw, J., Nathan, M. J., Ford, C. E., & Carnes, M. (2018). Low agreement among reviewers evaluating the same NIH grant applications. Proceedings of the National Academy of Sciences, 115(12), 2952-2957. https://doi.org/10.1073/pnas.1714379115

[3] Graves, N., Barnett, A. G., & Clarke, P. (2011). Funding grant proposals for scientific research: Retrospective analysis of scores by members of grant review panel. BMJ, 343, d4797. https://doi.org/10.1136/bmj.d4797

[4] TÜBİTAK (2020). ARDEB 1001 Programı: Proje değerlendirme sistemindeki yenilikler. https://tubitak.gov.tr/tr/duyuru/ardeb-1001-programi-proje-degerlendirme-sistemindeki-yenilikler

[5] European Commission (2024). Lump sum funding in Horizon Europe: What do I need to know?. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-funding-what-do-i-need-to-know_he_en.pdf